

where the mean and unbiased variance are estimated as:

$$\mathbb{R} \ni \hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n x_{i,j} \mathbb{I}(\mathcal{C}(x_i) = k), \quad (64)$$

$$\mathbb{R} \ni \hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{i=1}^n (x_{i,j} - \hat{\mu}_k)^2 \mathbb{I}(\mathcal{C}(x_i) = k), \quad (65)$$

where $x_{i,j}$ denotes the j -th feature of the i -th training instance. The prior can again be estimated using Eq. (32).

According to Eqs. (62) and (63), Gaussian naive Bayes is equivalent to QDA where the covariance matrices are *diagonal*, i.e., the off-diagonal of the covariance matrices are ignored. Therefore, we can say that QDA is more powerful than Gaussian naive Bayes because Gaussian naive Bayes is a simplified version of QDA. Moreover, it is obvious that Gaussian naive Bayes and QDA are equivalent for one dimensional data. Comparing to LDA, the Gaussian naive Bayes is equivalent to LDA if the covariance matrices are diagonal and they are all equal, i.e., $\sigma_1^2 = \dots = \sigma_{|C|}^2$; therefore, LDA and Gaussian naive Bayes have their own assumptions, one on the off-diagonal of covariance matrices and the other one on equality of the covariance matrices. As Gaussian naive Bayes has some level of optimality (Zhang, 2004), it becomes clear why LDA and QDA are such effective classifiers.

11. Relation to Likelihood Ratio Test

Consider two hypotheses for estimating some parameter, a null hypothesis H_0 and an alternative hypothesis H_A . The probability $\mathbb{P}(\text{reject } H_0 | H_0)$ is called type 1 error, false positive error, or false alarm error. The probability $\mathbb{P}(\text{accept } H_0 | H_A)$ is called type 2 error or false negative error. The $\mathbb{P}(\text{reject } H_0 | H_0)$ is also called *significance level*, while $1 - \mathbb{P}(\text{accept } H_0 | H_A) = \mathbb{P}(\text{reject } H_0 | H_A)$ is called *power*.

If $L(\theta_A)$ and $L(\theta_0)$ are the likelihoods (probabilities) for the alternative and null hypotheses, the likelihood ratio is:

$$\Lambda = \frac{L(\theta_A)}{L(\theta_0)} = \frac{f(\mathbf{x}; \theta_A)}{f(\mathbf{x}; \theta_0)}. \quad (66)$$

The Likelihood Ratio Test (LRT) (Casella & Berger, 2002) rejects the H_0 in favor of H_A if the likelihood ratio is greater than a threshold, i.e., $\Lambda \geq t$. The LRT is a very effective statistical test because according to the Neyman-Pearson lemma (Neyman & Pearson, 1933), it has the largest power among all statistical tests with the same significance level.

If the sample size is large, $n \rightarrow \infty$, and the θ_A is estimated using MLE, the logarithm of the likelihood ratio asymptotically has the distribution of χ^2 under the null hypothesis (White, 1984; Casella & Berger, 2002):

$$2 \ln(\Lambda) \overset{H_0}{\sim} \chi_{(df)}^2, \quad (67)$$

where the degree of freedom of χ^2 distribution is $df := \dim(H_A) - \dim(H_0)$ and $\dim(\cdot)$ is the number of unspecified parameters in the hypothesis.

There is a connection between LDA or QDA and the LRT (Lachenbruch & Goldstein, 1979). Recall Eq. (12) or (15) which can be restated as:

$$\frac{f_2(\mathbf{x}) \pi_2}{f_1(\mathbf{x}) \pi_1} = 1, \quad (68)$$

which is for the decision boundary. The Eq. (22) dealt with the difference of $f_2(x) \pi_2$ and $f_1(x) \pi_1$; however, here we are dealing with their ratio. Recall Fig. 1 where if we move x^* to the right and left, the ratio $f_2(x^*) \pi_2 / f_1(x^*) \pi_1$ decreases and increases, respectively, because the probabilities of the first and second class happening change. In other words, moving the x^* changes the significance level and power. Therefore, Eq. (68) can be used to have a statistical test where the posteriors are used in the ratio, as we also used posteriors in LDA and QDA. The null/alternative hypothesis can be considered to be the mean and covariance of the first/second class. In other words, the two hypotheses say that the point belongs to a specific class. Hence, if the ratio is larger than a value t , the instance $\mathbf{x}$ is estimated to belong to the second class; otherwise, the first class is chosen. According to Eq. (16), the Eq. (68) becomes:

$$\frac{(|\Sigma_2|)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^\top \Sigma_2^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\right) \pi_2}{(|\Sigma_1|)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^\top \Sigma_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \pi_1} \geq t, \quad (69)$$

for QDA. In LDA, the covariance matrices are equal, so:

$$\frac{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\right) \pi_2}{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \pi_1} \geq t. \quad (70)$$

As can be seen, changing the priors change impacts the ratio as expected. Moreover, the value of t can be chosen according to the desired significance level in the χ^2 distribution using the χ^2 table. The Eqs. (69) and (70) show the relation of LDA and QDA with LRT. As the LRT has the largest power (Neyman & Pearson, 1933), the effectiveness of LDA and QDA in classification is explained from a hypothesis testing point of view.

12. Simulations

In this section, we report some simulations which make the concepts of tutorial clearer by illustration.

12.1. Experiments with Equal Class Sample Sizes

We created a synthetic dataset of three classes each of which is a two dimensional Gaussian distribution. The means and covariance matrices of the three Gaussians from